Software Development Group Project

Project Proposal

SDGP Group : CS-124

Tutorial Group :11&14

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01) Problem

1.1 Introduction To The Problem

- In today's hectic and frequently stressful world, music has emerged as a popular emotional support system.Music is a potent tool for influencing our emotions. Whether it is for stress relief,mood enhancement,concentration or just relaxation. Even with their extensive collections and well selected playlists, Today's music streaming services are unable to adapt to the emotional demands of their consumers in real time. When users need immediate emotional relief or motivation, having to seek for songs or playlists that fit their mood might be annoying and useless.

- Although the majority of music apps provide generic playlists based on moods(such as "Happy Hills" or "Chill"), users must explicitly choose their desired mood,presuming they are aware of their exact needs.The Promise of real time emotional analysis - A method in which a music player may quickly recognize and react to user's emotional states, producing a customized listening experience without human input is disregarded by this procedure. Additionally competing systems are unable to give consistently relevant material since they do not take into account the small mood variations that happen during the same day.

- One of the biggest gaps in the music streaming market is the absence of an intelligent emotion responsive music player.By Seamlessly delivering the appropriate music at the exact right moment, A player that incorporates real time emotion detection could revolutionize the listening experience and improve user's well being. By seamlessly and intuitively satisfying customers' emotional demands this creative solution has the potential to completely transform the music streaming experience.

1.1.1. Problem Background and Stats

Problem Background

All around the globe,people are struggling with mental health issues, especially stress, anxiety, and depressive disorders. The fast pace of modern life, increasing work pressure, social deprivation and the curse of the digital explosion are responsible for this worrying trend. The COVID-19 pandemic only exacerbated matters and by December 2020, the worldwide prevalence of anxiety and depression increased by approximately 25%.

One of the areas that are impacted is the rising demand for emotional health management. Music, for instance, has been gaining ground as a most effective and convenient treatment. And since time immemorial, musical influence has been attributed such properties as mood change, stress alleviation and even mental health advancement.This happens due to the release of dopamine-indeed, how can it not be uplifting ? People listen to music when they must experience negative emotions and situations; Beyonce's song gives people strength when they are low , and calming music helps relieve agitation.

However, current music streaming platforms and devices do not fully leverage this therapeutic potential. While some apps offer mood-based playlists, these are typically static and require manual selection by the user. For individuals experiencing emotional lows or stress, manually selecting appropriate music may not be intuitive or convenient. This lack of real-time adaptability limits the accessibility of tailored musical support, which could provide immediate emotional relief for users. An AI-driven, emotion-responsive music player could bridge this gap, aligning music with the user’s current emotional state and offering a more personalized, supportive experience.

- Statistics on Emotional well-being:

- Research shows that music therapy can reduce the perception of pain by 21% to 25%in people with chronic conditions, including cancer and fibromyalgia. This is because music can activate parts of the brain related to pain modulation and emotional processing.
- The world health organization (WHO) reports that around 1 in 4 individuals will be affected by mental or neurological disorders at some point in their lives.
- According to a survey by the american psychological association, over 70% of adults report feeling stressed daily, often turning to music for relief
- Studies indicate that music can lower cortisol levels (the stress hormone) by 25%, helping reduce stress and anxiety
- Research has found that music therapy can reduce chronic pain, with participants in some studies reporting pain reduction by up 50% compared to controls groups. Music is also used to manage acute pain in medical procedures, with some studies suggesting reduced perception of pain during surgeries and procedures.

- Music and Emotion:

- A 2011 study by McGill University found that listening to music can increase dopamine levels by around 9% in areas of the brain related to emotional arousal and reward processing, such as the striatum. This effect helps explain why music can be so emotionally powerful and comforting, especially in times of stress.
- Emotions -synchronized music apps could help users in various moods and situations, but the lack of advanced AI-driven Solution limits the accessibility of tailored musical therapy.

1.1.2 Examples In The Problem

Example Scenario 1 : Anxious User

- Music that is calming or therapeutic may help a person who is stressed. In order to assist the user in relaxing, An Emotion Based Music Player can identify symptoms of stress and play soothing music,such as ambient sounds or slow instrumental pieces.

Example Scenario 2: Invigorated User

- Fast Paced, Energetic songs that fits their mood may be preferred by a person who is extremely energized. To keep their optimistic vibes going, The player could play fast paced music or music for working out.

Example Scenario 3: Saddened User

- If a person is depressed,they might want to listen to slower,more introspective music that expresses their emotions. Playing calming or gloomy music might make the user feel understood and help them process their feelings.

Example Scenario 4:Contented User

- Songs that praise and magnify happiness can be what someone in a good mood wants. Happy uplifting music might be played by the player to encourage the user to fully appreciate their mood.

Example Scenario 5: Nervous User

- For a person who is anxious,relaxing music may be necessary to reduce stress. The player may identify symptoms of anxiousness and play soothing sounds such as rain or ocean waves, or meditation music to help them unwind.

Example Scenario 6: A User With Nostalgia

- Songs from particular eras or genres may appeal to a user who is experiencing nostalgia. To evoke pleasant recollections, An Emotion Based Music Player might play some songs from the user’s past favorite eras by accessing their search history.

Example Scenario 7: Disinterested User

- A bored person might desire to listen to unusual or fresh music. To enhance excitement and diversity, The music player might present the user to songs by up and coming musicians or less well known songs.

Example Scenario 8: User Concentration

- The music player could determine the user's mood for concentration and generate instrumental, low lyric music that helps focus without being annoying to the user.

1.1.3 Attempted Solutions Of The Competitors

1)Spotify:

- Although Spotify provides activity based playlists like “Focus” and “Workout”,It does not offer real time emotion based changes. Playlists are still chosen by users by hand using mood keywords.

2)Youtube Music:

- Makes playlists recommendations based on time of day or particular activities, such as “Relax” although it doesn't instantly adapt to the user's emotional states. Mood based content selection requires user participation.

3)Amazon Music:

- Despite offering mood based playlists like “Happy Hills” and “Chill Vibes” Amazon Music lacks AI driven emotion detection and needs the user to choose them manually.

4)Moodagent:

- Although it lacks real time emotion identification capabilities and necessitates continuous user engagement,Moodagent lets users manually select their mood. Such as joyful or sad.

5)Calm App:

- Mostly used for meditation and relaxation.Calm provides soundscapes and music to help users unwind. Although it doesn't have a real time playlist adaptation based on emotions, It does offer music with a concentrate or relaxation theme.

6)Deezer:

- Does not have automatic emotion detection,however it does have some mood playlists. With no AI help with real time modifications, Users are forced to pursue mood based content.

7)Apple Music:

- Although it provides carefully chosen mood playlists like “Feel Good” and “Chilli” Apple Music does not have a real time emotional reactivity. There is no AI driven emotion recognition or mood adaptation in the app, Instead users must manually choose playlists based on mood keywords.

Feature Comparison Chart

Feature	Spotify	Apple Music	Youtube Music	Amazon Music	Mood Agent	Calm App	Deezer	Our Emotion Based Music Player
Activity Based Playlists	✓	✓	✓	✓	✓	✗	✓	✓
Manual Mood Selection	✓	✓	✓	✓	✓	✓	✓	✓
Real Time Emotion Detection	✗	✗	✗	✗	✗	✗	✗	✓
Automated Emotion Based Playlists	✗	✗	✗	✗	✗	✗	✗	✓
Mood Adaptive Transitions	✗	✗	✗	✗	✗	✗	✗	✓

02) Proposed Solution

- Nowadays depression and anxiety is a vastly spreading mental problem that may cause severe damages to the society such as suiciding, mental unsatisfaction, mood swings, suffering from suicidal thoughts as well as unbearable sadness. These problems can be the cause of society's downfall as well as increase of the death counts of suiciding. So to prevent these types of imbalance/downfalls there are many solutions that might be useful such as AI powered chatbots, Yoga, therapists, psychiatrists, personal doctors, mental health camps, counselors who can be hired to listen to their problems. With the vast development of technology nowadays most of the people have mobile phones or some type of a digital device at least one device per family. So we thought of coming up with a digital solution. Also we thought of creating an application that can run on any device.

- So as a proper solution we came up with an application to fix the user's current mood. Further details about our applications we created this application to scan the user's biometrics, facial reaction as well as user's voice tone to know the current mood. And after scanning the current mood, if the current mood of the user is sad the application will suggest some songs/tunes to convert the user's bad mood to a better/good mood.

- As a proper solution for running this application on any device we mainly focused on mobile phones, laptops as well as computers so the user can run this application on either device to make himself better and happy. Not only for making the mood, our application can be used for listening to music without any hesitation.

03) Target Audience

3.1 Introduction for Target Audience.

Identifying our target audience in the music industry is a crucial aspect of effective music marketing. Understanding the target market is essential while developing a music app to produce a product that appeals to customers requirements and listening tastes. From devoted fans and casual listeners to people who depend on music for particular tasks like concentration or health, music apps serve a wide range of users. App developers can prioritize functionally, design, and marketing tactics that appeal to various user types by identifying the target demographic.

3.2 Types of Target Audience for the Music app.

- **Global Listeners with Culturally Diverse Tastes.**

Listeners who love exploring global music would find an Emotion Tracking Music App especially appealing, as it offers emotion-specific playlists from diverse cultures. Through mood-based music recommendations, the app allows users to explore new genres, languages, and cultural sounds while also finding songs that speak to their emotions . This special fusion of diversity and personalization offers a deeper, more engaging listening experience, introducing users to music that expands their musical horizons and expresses their emotions.

- **Young Adults and Teenagers.**

Mood tracking technology appeals to younger, digitally native consumers because it allows them to analyze emotions, reduce stress, and learn how to control mood swings. They like interactive features like daily mood insights and music history-based recommendations, as well as customized playlists that adjust to their emotional fluctuations. They can connect with others through similar musical mood and discoveries thanks to social sharing tools, which further improve the experience. A dynamic, captivating platform that satisfies their social and emotional demands is produced by combining personalization with emotional support.

- **Fitness Enthusiasts and Athletes.**

Fitness enthusiasts, particularly those in sports that require high focus and sustained motivation, would find great value in an emotion-based music app tailored to their workout needs. Whether they're looking for energizing tracks to boost motivation during a challenging run or calming music to aid post-workout recovery, these users benefit from playlists that align with both workout intensity and their current mood.

- **Emotional Health Advocates**

A mood-tracking music app that supports their journey might be beneficial for people who are focused on their emotional well-being, such as those in treatment. Users can learn more about how music affects their mental health with tools like emotion-based music history and mood journaling. Its function as a therapeutic tool is further enhanced by integrations with wellness initiatives or mental health resources, which provide assistance for introspection and emotional development.

04) Resource Requirements

This section covers the key resources for creating an emotional based music player, including the Hardware Requirements, Software Requirements and Technology Stack.

4.1 Hardware Requirements

Depending on the system's size, Target Audience and emotional analysis difficulties, various hardware is needed to build and implement an emotional based music player and generator. The following Hardware could significantly improve performance and features.

1. **High-performance Workstations:** Development workstations should need multi-core processors (Intel i7/i9) and at least 16GB RAM to support run hard programs like music generating software and Machine Learning Models.

2. **GPU (Graphics Processing Unit):** If the application mainly uses deep machine learning models like CNNs and RNNs for emotion recognition, GPU acceleration could be necessary.

3. **High-Resolution:** High-Resolution setups improve development tasks and productivity.

4.2 Software Requirements

A wide range of Software tools and Systems are required to ensure that the Emotional Based Music Player is developed, set up, and maintained properly.

★ **Development Environment Tools**

- **IDE (Integrated Development Environment):**
  - Visual Studio Code (VS Code)
  - PyCharm
  - Intellij IDEA
  - Flutter

These are some tools essential for code development and testing.

- **Version Control System:**
  Version Control is essential for collaborative development for tracking code changes over time and the systems like GitHub provide it.

★ **Operating Systems**

- Windows
- Linux
- macOS

A development environment can be included in any of these major operating systems.

★ **DBMS (Database Management System)**

Relational Databases like MySQL could be used to store structured information like user profiles, Choices and use data.

★ **Machine Learning Frameworks**

- **OpenCV**: Expect to use for face emotion identification and image processing and especially for identifying emotions from facial expressions.
- **PyTorch:** Expect to use for creating, developing and implementing emotion identification models and neural networks.

★ **Music Generating Software**

- **Generative Adversarial Networks (GANs) for music Generation**: Might be used to create original creative musical arrangements.

- **Librosa:** A Python library for helping tasks like tempo and beat analysis, in audio analysis.

- **MIDI Processing Softwares (Musical Instrument Digital Interface):** These Softwares can be used when the system controls MIDI files.

4.3 Technology Stack

An emotion based music player's Technological stack includes a combination of front-end, back-end and data processing technologies. Then following Stack could be used.

★ **Front-end**

- **React.js :** Creating adaptable and dynamic user interfaces with features like mood-based routing, playlist management is easy by JavaScript frameworks like React.
- **Bootstrap / Tailwind CSS:** CSS frameworks for style , enabling programmers to design an innovative, user-friendly interface that works on a variety of devices.

★ **Back-end**

- **Node.js / Django:** Creating the main application and managing user interactions with the database are handled by server-side frameworks like Node.js or Django.

★ **Machine learning and Data processing**

- **Keras:** Expect to use for fast developing and testing deep learning models for music production and emotion recognition.

★ **Database**

- **MySQL:** This relational database system uses structured data, such as user profiles, playlist details and past musical tastes.
- **MongoDB:** Prefer to manage emotional data, music files and user data in NoSQL databases.

★ **Cloud**

- **Google Cloud / Microsoft Azure:** Large datasets and machine learning models may be managed in a flexible and economical way thanks to cloud services.

05) Features of the solution

This Emotion-based Music Player concept is designed to deliver a highly personalized music experiences by interacting users' emotions through advances technologies and offering AL-generated music and visuals that align with This Emotion-Based music Player concept is designed to deliver a highly personalized music experience by interpreting users' emotions through advanced technologies and offering AL-generated music and visuals that align the detected mood.

5.1 Facial and voice Recognition for Emotion Detection

The system uses facial recognition and voice analysis of the user's emotions. By detecting facial expressions and voice tones, the player can identify whether the user feels happy, sad,calm or other moods. This knowledge guides the selection of music, which corresponds with the identified emotional state.

5.2 Generative Al Feature to create songs Based on Emotional Mood

After determining the user's mood and emotions, a machine learning system composes an audio track that perfectly captures the essence of the moment. For example, the AL might create a calming instrumental song if the user seems calm,or an exuberant pop song if they are feeling energized.

5.3 Option to Generate songs based in user preferences

Users can choose their preferred genre of music manually and avoid emotion detection. They can select from a variety of genres,moods,or topics and the AI will compose a song based on their tastes,giving them complete control over their musical selection.

5.4 Al Image Generation to Match the Song

An Al image generator first generates a song, then produces an image that visually reflects the song's atmosphere. This image could be an abstract depiction of the mood of the song, giving a visual component that complements the sound to improve the whole experience.

5.5 Popup Questions for Better Analysis

Pop-up questions are incorporated into the system to help it better grasp the user's preferences and emotions. These questions may focus on present emotions, preferred genres, or planned activities. These inputs allow the Al to respond in a more customized way,Increasing the created music's relevance.

5.6 Surprise Me Option for Music

The "Surprise Me" option creates random music and images for users who are receptive to unplanned encounters without depending on user input or mood recognition. For individuals who prefer diversity,this function keeps the listening experience novel and exploratory